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TEAM NAME

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MEMBERS

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THEME

BioTech & HealthTech

PROBLEM STATEMENT

AI-Assisted Triage for Primary Health Centres-

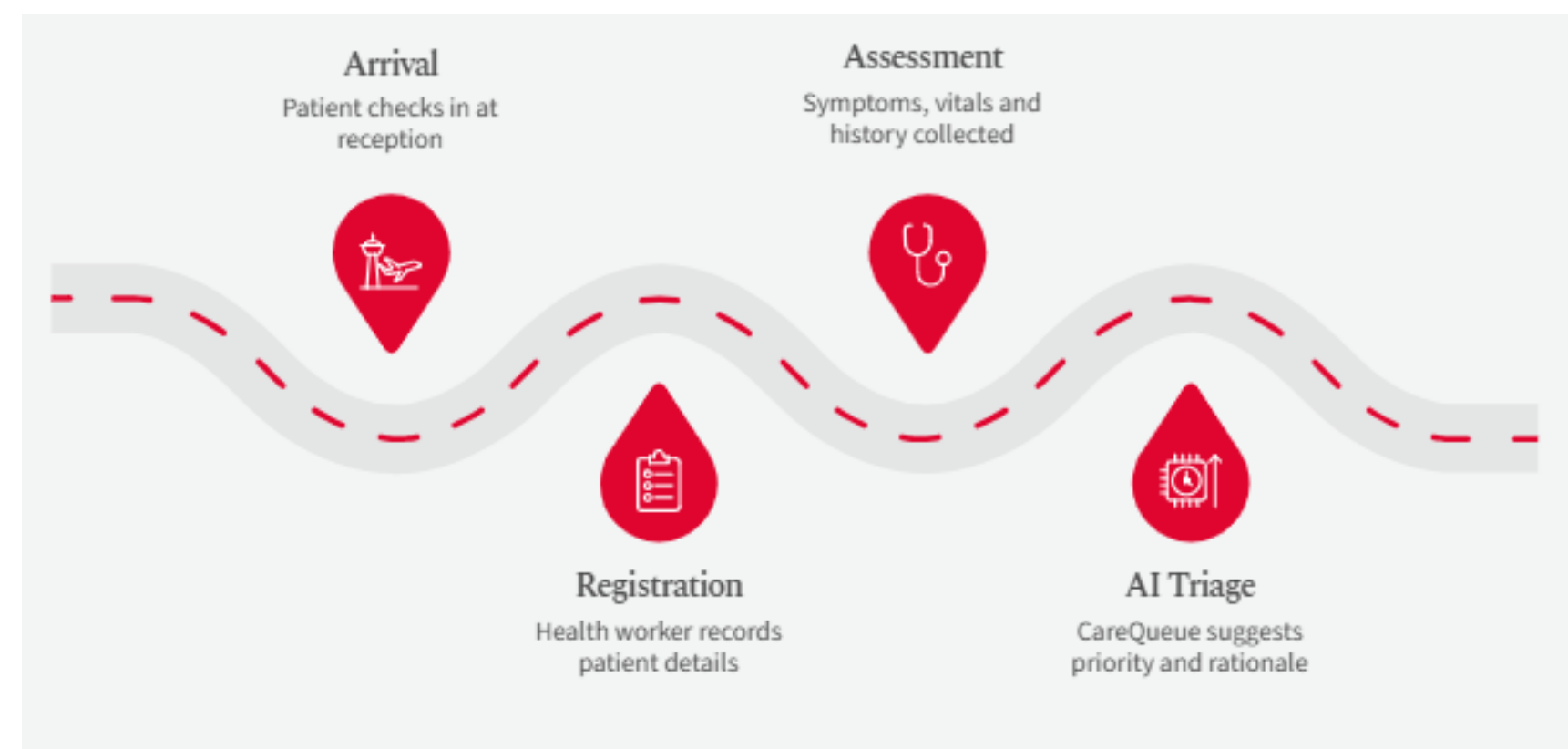
Rural primary health centres often have one doctor serving thousands of patients, leading to delayed identification of serious cases. Create a triage support app that helps health workers prioritize patients based on symptom severity.

Proposed Solution — CareQueue AI

When several patients require attention, the doctor needs a structured way to decide who should be assessed first, especially when urgency isn't obvious from appearance alone. That's when we need CareQueue AI.

KEY FEATURES

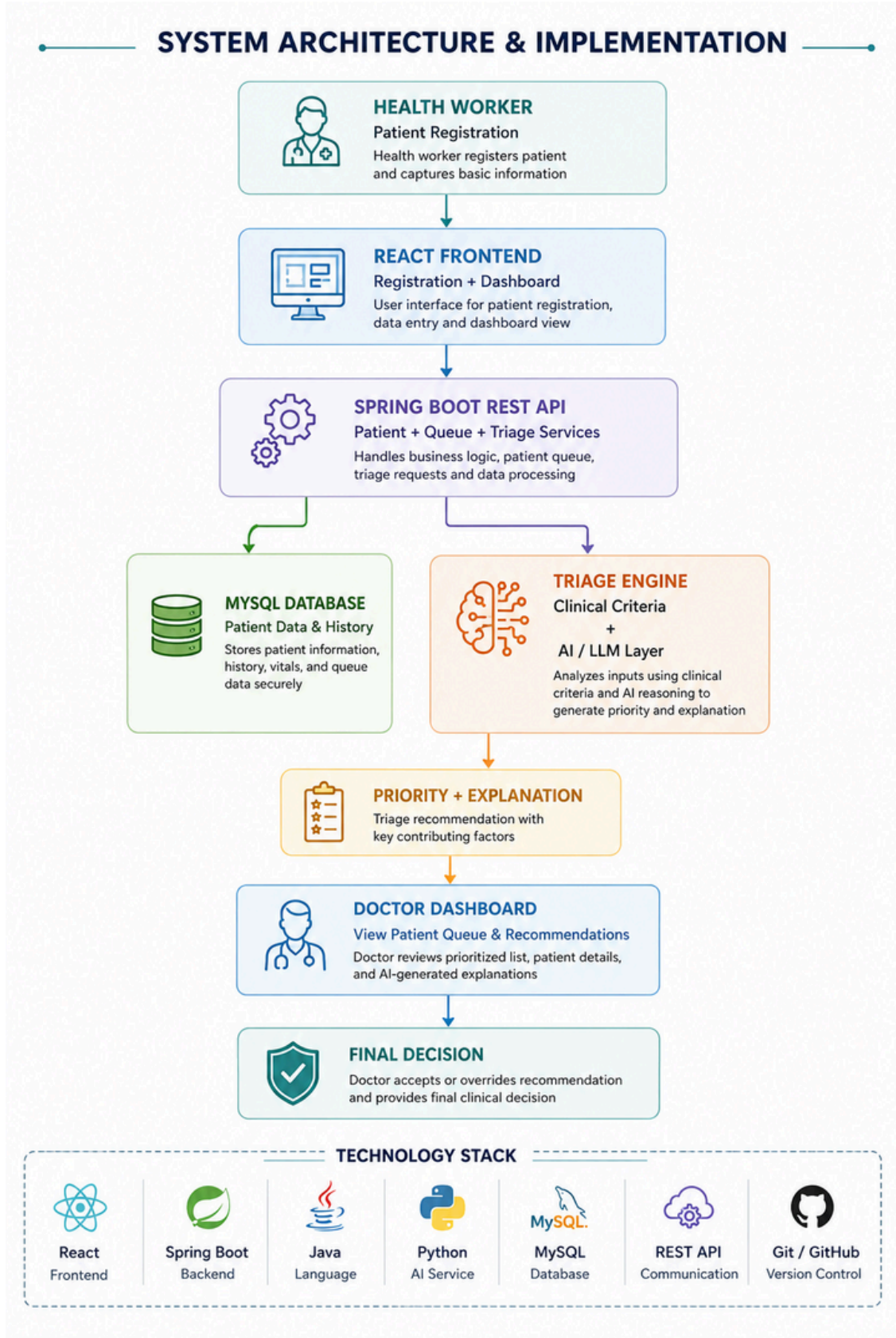
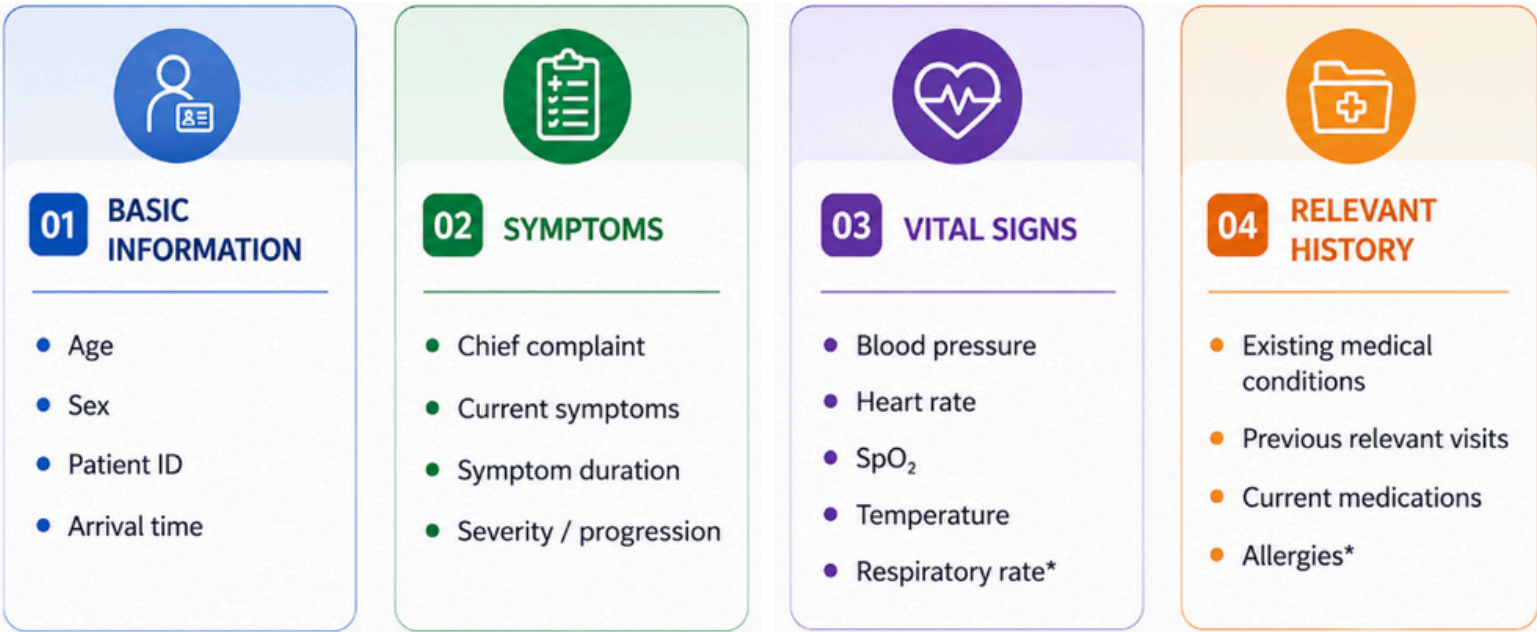
HOW IT WORKS



 Patient Registration Health worker quickly records basic patient information upon arrival.	 Symptom & Vital Input Captures symptoms and available vitals: temperature, heart rate, blood pressure, SpO2.
 Doctor Dashboard Structured view of patients with relevant information for clinical assessment.	
 AI-Assisted Triage Analyzes inputs using clinical triage criteria and an AI reasoning layer.	 Priority + Explanation Provides a priority recommendation and highlights key contributing factors.
 Doctor Override Doctor can accept or override the AI recommendation. Final decision always with the doctor.	

Innovation & Uniqueness

- **Explainable Triage** Not just a priority label — the system highlights important contributing factors.
- **Human-in-the-Loop**
AI recommends; the doctor can accept or override the recommendation.
- **Patient Context**
The doctor receives relevant symptoms, vitals and patient information alongside the recommendation.
- **Pre-Visit Symptom Capture**
Patients record their concerns beforehand, reducing communication gaps and saving consultation time.



 **CAREQUEUE AI**
Doctor Priority Dashboard

≡ Patient Queue

3 waiting

PATIENT	AGE	PRIORITY	WAITING
P4 P-104 SELECTED	65	HIGH	8 min
P8 P-108	42	MEDIUM	15 min
P2 P-112	31	LOW	22 min

Priority order is a proposed suggestion for review. Ranking does not represent a clinical diagnosis.

 Patient Details

SELECTED PATIENT

P-104

P4

♡ SYMPTOMS

✓

Difficulty breathing

🔻 SPO₂

88%

📶 HEART RATE

110 bpm

☆ AI-ASSISTED RECOMMENDATION

HIGH PRIORITY

Key contributing factors

• Low SpO₂

• Difficulty breathing

Prototype output only — for concept demonstration, not a clinical diagnosis or validated recommendation.

ACCEPT RECOMMENDATION

OVERRIDE

Feasibility

- Software-first: React, Spring Boot, Python & MySQL
- No specialized hardware for the initial prototype
- Modular AI layer allows model flexibility
- MVP: Patient Input → Triage → Priority → Doctor Review

Challenges & Risks

- Clinical reliability: Incorrect AI recommendations
- Data availability: Limited clinical/triage data
- Privacy: Sensitive patient information
- Connectivity: Unreliable rural internet

Mitigation

- Reliability → Clinical criteria + AI + doctor override
- Data → Public/simulated/reference cases
- Privacy → Authentication + secure data handling
- Usability → Simple forms + predefined symptoms
- Connectivity → Low-bandwidth design; offline mode as future scope

Patients

- Earlier identification of potentially urgent cases
- Pre-visit symptom capture reduces communication gaps

Doctors

- Structured patient view during rounds
- Faster access to relevant patient information

Healthcare Workers

- Standardized triage support
- Simple symptom & vital collection

Healthcare System

- More structured patient prioritization
- Scalable across healthcare centres

- [1] WHO — Emergency Triage Assessment and Treatment (ETAT), 2005.
- [2] Han & Choi — LLM-based Multi-Agent CDSS for KTAS Triage, 2024.
- [3] Korean Society of Emergency Medicine — Korean Triage and Acuity Scale (KTAS).
- [4] WHO — Ethics and Governance of Artificial Intelligence for Health, 2021.
- [5] Han & Choi — Multi-Agent CDSS for KTAS-Based Triage, AAIML, 2025.